

Born in Simulation: How Visual Domain Randomization Bridges the Sim-to-Real Gap for Object Recognition

An Empirical Study of Colour Diversity and Learned Representations

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Abstract

Physical AI systems cannot be trained safely or affordably in the real world, so they are increasingly trained inside simulation. This creates the *sim-to-real gap*: a model that performs well in simulation often fails on real sensor data. Visual domain randomization, introduced by Tobin et al. (2017), addresses this gap by deliberately varying the appearance of the simulation so the model learns transferable structure rather than simulation-specific shortcuts. We present an independent small-scale study of this principle. Six ImageNet-pretrained ResNet-18 classifiers are fine-tuned entirely on synthetic images of five YCB objects rendered in PyBullet, each at an increasing level of visual diversity, and every model is evaluated on real photographs it never saw during training. Diversity sharply improves real-world accuracy — from 8.6% with no randomization to a peak of 61.3% — but the relationship is *non-monotonic*, with an optimal level beyond which the highest-diversity model begins to overfit. Grad-CAM analysis reveals the mechanism: models trained without randomization attend to the background, while models trained with sufficient diversity attend to object geometry. An ablation without pretraining further shows that, at our data scale, ImageNet pretraining remains essential, locating a scale boundary below Tobin et al.’s regime.

Keywords

domain randomization, sim-to-real transfer, Physical AI, object recognition, synthetic data, Grad-CAM, YCB dataset

1 Introduction

Physical AI — the branch of artificial intelligence concerned with agents that perceive and act in the physical world — faces a fundamental deployment problem. The algorithms that power perception and control require enormous quantities of training data, yet collecting that data on real robots is slow, expensive, and often dangerous. A robot learning by trial and error in the real world may damage itself or take months to gather enough experience. Simulation offers an escape: a virtual environment can generate millions of training examples cheaply, safely, and in parallel. Yet a simulator cannot reproduce reality exactly, because modelling the world to the fidelity of real physics and optics is computationally intractable; every simulator therefore takes shortcuts in lighting, texture, physics, and sensor behaviour. These shortcuts make simulated data systematically different from real data, and a model trained purely in simulation frequently fails in the real world. This discrepancy is the *sim-to-real gap*, and bridging it is one of the central challenges of modern Physical AI.

The intuitive remedy is to make the simulation more realistic. But photorealism is expensive to author and compute, and some residual gap always remains. Tobin et al. (2017) proposed the opposite, counterintuitive move, which they called *visual domain randomization* [5]. Rather than making the simulation look more real, they made it deliberately varied — randomizing object textures, background textures, lighting, and camera position at every training step — on the hypothesis that if the variability in simulation is wide enough, the real world becomes just another sample from a distribution the model has already seen. Their results were striking: a network trained entirely on non-realistic randomized simulation achieved 1.5 cm object-localization accuracy in the real world with zero real training data, and their ablation showed the *diversity* of textures was the single most important factor driving transfer.

This paper asks whether that core finding holds in an independent, small-scale reproduction using open-source tools, and probes a question the original ablation raised but did not fully resolve: *how much* diversity is needed, and what changes inside the network as diversity grows. Concretely, we investigate the following research question:

How does the diversity of visual domain randomization affect sim-to-real transfer accuracy for object recognition — is there a threshold beyond which additional diversity yields diminishing returns — and what changes in the neural network’s learned representations as diversity increases?

We answer this in two complementary parts. A *quantitative* part trains six classifiers at increasing diversity levels and measures real-world accuracy, revealing a non-monotonic relationship with a clear optimum. A *qualitative* part uses Grad-CAM to visualize what each network attends to, providing a mechanistic explanation for that trend. An ablation without pretraining then probes whether diversity can substitute for ImageNet pretraining at our data scale. Taken together, our contribution is not a single accuracy number but a characterization of *how* and *why* visual diversity governs transfer, and of the boundaries within which the principle holds.

2 Background and Related Work

2.1 The Reality Gap

The differences that separate a simulator from the world it models arise from several sources. Tobin et al. identify imperfect system identification, unmodeled physical effects, low-fidelity sensor simulation, and rendering limitations [5]. For a purely visual perception task the rendering gap dominates: the simulated camera image simply does not look like a real photograph. Domain randomization

attacks this gap not by closing it directly but by making the model robust to it. If the network has seen thousands of visually different versions of an object during training, the particular appearance of the real object is no longer foreign — it falls inside the distribution the model already covers.

2.2 Domain Randomization, and Why It Is Not Augmentation

A common objection is that domain randomization is merely a form of image augmentation. The distinction is important. Image augmentation applies pixel-level transforms — flips, crops, brightness shifts — to existing real photographs; the underlying data, and therefore the domain, is still reality. Domain randomization instead generates training data entirely within a simulator, varying the scene *before* rendering: object pose, camera viewpoint, lighting, distractor objects, and surface appearance are all sampled anew, and each configuration is rendered from scratch. Varying the camera viewpoint in three dimensions and inserting genuine 3D distractor objects are operations that 2D augmentation cannot perform, because they require a geometric model of the scene. The goal differs too: augmentation makes a model robust to nuisance variation in real data, whereas domain randomization aims to make simulation so varied that reality becomes an unremarkable sample.

2.3 Extensions and Limits

The principle was scaled dramatically by OpenAI’s Dactyl system, which trained a 24-degree-of-freedom robot hand to solve a Rubik’s cube entirely in simulation by combining visual, dynamics, and sensor randomization, and introduced Automatic Domain Randomization to widen ranges adaptively [3]. Dactyl confirmed that randomization scales to complex manipulation, but also showed that visual randomization alone is insufficient for contact-rich physical tasks — dynamics and sensor gaps must also be addressed. Our study isolates the visual component, which is the dominant factor for the pure recognition task we consider.

2.4 The YCB Object Set

We draw our objects from the Yale-CMU-Berkeley (YCB) object and model set [1], a standard manipulation benchmark that provides both high-resolution real photographs and scanned 3D meshes for each object. This dual availability is central to our design: the meshes drive our simulation, while the photographs serve as a held-out real-world test set, so that the model is tested on the same objects it trained on but never on images that resemble its training data.

3 Methodology

3.1 Task and Data

We frame the problem as five-class object recognition. Given a 224×224 image, the model predicts which of five YCB objects it contains: a mug, a mustard bottle, a power drill, a bowl, and a bleach cleanser. These five were chosen for shape diversity — cylindrical, rectangular, asymmetric, flat-curved, and tall-cylindrical respectively — so that any observed effect is not tied to a single geometry.

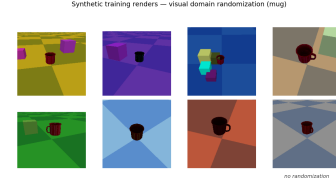


Figure 1: Example synthetic training images. Each render varies object colour, floor colour, camera pose, and distractor objects; the same mesh appears under widely different visual conditions. Because the camera pose and 3D distractors are sampled per scene, this is domain randomization, not 2D augmentation.

Every model is trained exclusively on synthetic images and evaluated exclusively on real YCB photographs, with 600 real images per class forming the 3,000-image test set. No real image is ever seen during training.

3.2 Simulation and Randomization

Synthetic images are rendered in PyBullet in headless mode. For each image the renderer loads the object’s 16k mesh, places it on a ground plane, and — when randomization is enabled — assigns a random colour to the floor and to the object, samples a random camera distance (0.35–0.75 m), yaw (0–360°), and pitch (−70° to −15°), and scatters zero to four randomly coloured distractor boxes into the scene. The distractors follow Tobin et al. directly: their purpose is to prevent the network from assuming the largest or most central object is always the target. Because each image is a fresh render of a re-sampled 3D scene rather than a transform of a fixed photograph, this pipeline is domain randomization in the sense developed in Section 2.2, not augmentation.

3.3 Colour versus Texture Randomization

An honest limitation concerns what we randomize. Tobin et al. randomized *photographic textures* — images of wood, concrete, or fabric — mapped onto object surfaces. PyBullet’s built-in `ER_TINY_RENDERER` does not support arbitrary texture-image mapping; the only available handle on appearance is the RGBA colour value, applied as a multiplicative tint over the mesh. Our diversity levels therefore correspond to the number of unique *colour* combinations rather than distinct texture images, and we use the term *colour randomization* to describe this faithfully. The hypothesis under test — that greater visual variety during simulation training improves real-world transfer — is unchanged, but our absolute numbers should be read as a simplification rather than a direct numerical comparison with the original work.

3.4 Model and Training

All models share an identical architecture: ResNet-18 [2] pretrained on ImageNet, with its final fully-connected layer replaced by a fresh Linear(512 → 5) head. The entire network is then fine-tuned; no layers are frozen, so this is full fine-tuning rather than feature extraction. Pretraining matters because, with only a few thousand

synthetic images per model, learning low-level features from random initialization is infeasible; the ImageNet weights supply general edge and part detectors that fine-tuning specializes to our five classes. Training uses cross-entropy loss, the Adam optimizer at a learning rate of 1×10^{-4} , a batch size of 32, and early stopping (patience 12, maximum 100 epochs) against a 15% stratified per-class validation split. These settings are held identical across all conditions, so that the only variable that changes between models is the diversity of the training data — the definition of a controlled experiment.

3.5 Experimental Conditions

We train one model per diversity level: a no-randomization baseline (fixed floor, fixed object tint, fixed camera) and five randomized levels of 10, 100, 500, 1,000, and 5,000 unique colour combinations. Diversity is controlled by the seed-cycling scheme of Eq. (1), so that exactly *level* unique scenes are produced per class. Our primary experiment uses 5,000 images per class (150,000 synthetic images in total); an earlier run at 1,000 images per class confirms robustness, and a still earlier run is discussed in Section 4 as the source of a methodological correction.

4 A Note on Experimental Validity

Diversity is controlled by a seed-cycling scheme. For image i of an object, the scene parameters are drawn from a seed

$$s(i) = s_0 + (\text{obj} \times C_{\text{obj}}) + (\text{level} \times C_{\text{lev}}) + (i \bmod \text{level}), \quad (1)$$

where the $i \bmod \text{level}$ term cycles through exactly *level* distinct seeds, and the object and level offsets guarantee a disjoint seed range for every object and every level. Our first run used a simpler rule without the level offset, $s(i) = s_0 + (i \bmod \text{level})$. With 500 images per class, any level of 500 or greater satisfies $i \bmod \text{level} = i$ for every $i < 500$, so the 500, 1,000, and 5,000 levels drew from the same seeds and generated *identical* training data — that run contained only four distinct datasets rather than six, and the apparent differences among the three highest levels reflected training variance, not diversity. We corrected the rule to Eq. (1) and increased the data to 5,000 images per class, so that every level now produces a genuinely distinct, non-repeating set of scenes up to its nominal count. We report this openly because the discovery and correction of such errors is part of sound experimental practice.

5 Results

5.1 Accuracy versus Diversity

Figure 2 reports overall top-1 accuracy on the real test set. Three findings stand out. First, randomization works dramatically: the no-randomization baseline reaches only 8.6% — below the 20% chance level for five classes — while moderate diversity lifts accuracy several fold. Second, accuracy peaks at the 1,000 level at 61.3%, a clear optimum. Third, and most striking, the curve is *non-monotonic*: the 5,000 level falls back to 37.6% even though it contains the most unique scenes of any condition. More diversity is not uniformly better — past an optimum it actively hurts. Inspection of training reveals why: the 5,000 model is the only one whose validation loss rose while its training loss kept falling — it overfits the synthetic distribution, and early stopping halts it early. The intermediate

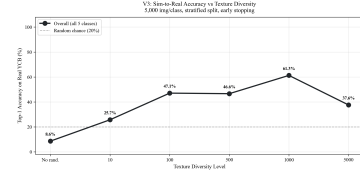


Figure 2: Real-world accuracy as a function of colour diversity. Accuracy rises steeply from the below-chance baseline, peaks at the 1,000 level (61.3%), and declines at 5,000 as that model overfits. The dashed line marks the 20% five-class chance level.

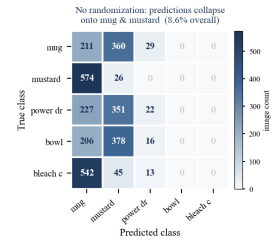


Figure 3: Confusion matrix for the no-randomization model (rows: true class, columns: predicted). Predictions collapse onto two classes, confirming that the below-chance accuracy reflects a confident but wrong shortcut rather than random guessing.

1,000-images-per-class run peaks at the same level (65.3%), confirming the optimum is robust rather than a single-run artifact.

That the baseline scores *below* chance deserves comment, because a naive reading suggests something broken. The confusion matrix (Figure 3) shows the opposite of randomness: the no-randomization model collapses onto two classes, predicting almost every test image as mug or mustard bottle. It is not guessing — it is confidently applying a shortcut learned in simulation that does not hold in reality, and so it is reliably wrong rather than uniformly uncertain. This is exactly what domain randomization is meant to prevent.

5.2 Per-Class Behaviour

Per-class accuracy reveals the structure behind the aggregate trend. The most dramatic case is the bowl, whose flat circular profile is geometrically ambiguous across the wide range of camera pitches used in simulation. The bowl was nearly unrecognizable in earlier runs, yet at the 1,000 level it reaches 76.7% — learnable only once the data is both genuinely diverse and trained to convergence. The power drill, whose elongated asymmetric shape is the most distinctive, is recognized most reliably across levels. Other classes are less stable: the bleach cleanser peaks at low diversity and declines as diversity grows, and the mug remains weak throughout. The honest summary is that diversity improves *aggregate* transfer up to an optimum, while *which* class benefits at *which* level is unstable — itself evidence that the specific scenes generated, not merely their count, shape what is learned.

5.3 What the Network Learns

Grad-CAM [4] heatmaps computed at the final convolutional block make the mechanism visible (Figure 4). For every object tested, the no-randomization model attends to the background — the wall behind the mug, the floor beneath the mustard bottle — and predicts incorrectly, while the best diversity model shifts its attention onto the object itself and predicts correctly. The clearest comparison is the bleach cleanser: the baseline scatters weak activation across the lower frame and predicts mug, whereas a diverse model concentrates a tight hotspot on the bottle and classifies it correctly. This is direct visual evidence that randomization shifts the network’s learned attention from background context to object structure — the qualitative half of our research question.

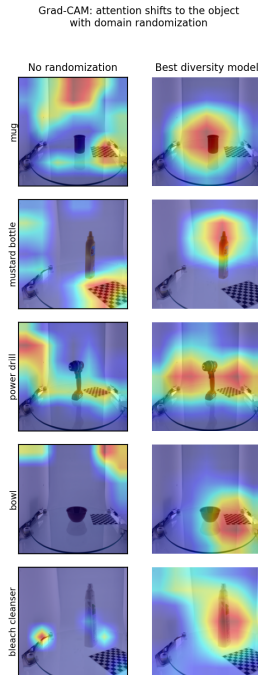


Figure 4: Grad-CAM comparison. For each class the no-randomization model (left) and the best diversity model (right) are shown; warm regions indicate high activation. The baseline attends to background, the diverse model to the object.

5.4 Diversity versus Pretraining

To test whether diversity can substitute for ImageNet pretraining at our data scale, we retrain the baseline and the best (1,000) level from random initialization. The result is informative in two ways. At the no-randomization level, the randomly-initialized model collapses entirely, predicting a single class for all 3,000 test images and so scoring exactly at the 20% chance level — a degenerate constant predictor, not learning. At the 1,000 level this variant does genuinely learn — its predictions spread across all five classes — but reaches only 25.0%, far below the 61.3% of its pretrained counterpart. At our scale, then, diversity alone does not replace pretraining: the

synthetic data teaches the network how to *recombine* pretrained features, but not how to learn good visual features from nothing. This locates a scale boundary beneath the regime in which Tobin et al. found pretraining unnecessary, and identifies data volume as the variable that separates the two.

6 Discussion

The non-monotonic curve is the most surprising outcome. The decline at 5,000 has a concrete explanation visible in training: the highest-diversity model is the only one whose validation loss rose while its training loss continued to fall, overfitting the synthetic distribution before early stopping halted it. Below that point, a plausible account of the dip-then-peak shape is the interaction between diversity and exposure under a fixed budget — at low diversity each scene is seen many times and a few associations are strongly reinforced, while intermediate diversity spreads exposure thin before there is enough variety to compensate. Either way, the result refines Tobin et al.’s ablation, which treated more texture diversity as monotonically beneficial.

The Grad-CAM evidence ties the quantitative and qualitative halves together. The baseline’s failure is not random but systematic — it learns a background shortcut that exists in simulation but not in reality, and the confusion matrix shows it collapsing onto a pair of classes as a result. Randomization removes that shortcut by making the background uninformative, forcing reliance on the one cue stable across renders: the object’s shape. The ablation without pretraining adds a boundary condition. Diversity helps the pretrained model dramatically; the randomly-initialized model improves only marginally, and even that small gain reflects genuine, if weak, learning rather than the degenerate collapse seen at its baseline. Only the pretrained model converts diversity into strong transfer at our data scale — the volume at which diversity can stand on its own lies above what we use here, and below the far larger regime of the original work.

7 Limitations

Three limitations bound our conclusions. First, as detailed in Section 3.3, we randomize colour rather than photographic texture, a simplification imposed by the renderer; a texture-mapping pipeline — sampling the Describable Textures Dataset within a renderer such as PyRender or Blender — would test the principle more faithfully, but was outside the time and compute budget of a one-semester seminar. Second, each model was trained once, without repeated runs, so we cannot place confidence intervals on the curve, and some non-monotonic behaviour may reflect training variance. Third, hyperparameters were fixed across all conditions, isolating diversity as the sole variable but leaving each level individually unoptimized. None of these undermines the central findings — that colour diversity sharply improves transfer, that the relationship is non-monotonic with an optimum, and that the mechanism is a shift from background to object attention.

8 Conclusion and Future Work

We set out to test, independently and at small scale, whether the diversity of visual domain randomization governs sim-to-real transfer, and to look inside the network for the reason. Our results confirm

Tobin et al.’s core claim in a new context: diversity lifts real-world accuracy from below chance to a 61.3% peak, and Grad-CAM shows it does so by redirecting attention from background to object geometry. We also find what the original ablation did not report — that the relationship is non-monotonic, that an optimal diversity level exists for a fixed training budget, that the highest-diversity model declines through overfitting rather than any shortage of variety, and that pretraining remains essential at our data scale. Future work should repeat each condition to quantify variance, probe why flat objects such as the bowl resist recognition from simulation, and combine photorealistic rendering with randomization — varying realistic textures rather than flat colours — to test whether realism and diversity are complementary rather than opposed. Taken together, the study reinforces a central lesson of Physical AI: the path across the reality gap runs not through more realistic simulation, but through more varied simulation — in the right amount.

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